

STATISTICS 652

LECTURE 5 - 26

REVIEW PROPERTIES (FROM 593)

- ANY FUNCTION OF A COMPLETE STATISTIC IS COMPLETE.
- ANY 1-1 FUNCTION OF A MINIMAL SUFFICIENT STATISTIC IS MINIMAL SUFFICIENT.

PROOF: HW

EXAMPLE: $X_1, \dots, X_n \text{ iid } N(\mu, \sigma^2)$ $\mu \in \mathbb{R}, \sigma^2 > 0$ with $n \geq 2$

$\Rightarrow (\bar{X}, S^2)$ ARE MINIMAL SUFFICIENT AND COMPLETE (WHY?)

HENCE, $\bar{X}, S^2$ ARE UNIQUE UMVUE FOR μ, σ^2 RESPECTIVELY

NOTATION: $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

EXAMPLE: FOR $n=1$, i.e. $X \sim N(\mu, \sigma^2)$

$(T_1, T_2) = (X, X^2)$ IS MINIMAL SUFFICIENT AND COMPLETE.

It is complete since we have a FULL RANK 2-PARAMETER EXP. FAMILY.

NOTE: $E[g(T_1, T_2)] = E[T_1^2 - T_2] = 0$ WITH $g(t_1, t_2) \neq 0$.

THIS DOES NOT CONTRADICT COMPLETENESS SINCE

$g(t_1, t_2) = t_1^2 - t_2 = 0$ ON THE SUPPORT OF (T_1, T_2) ,

SO, UNIQUE UMVUE FOR μ IS $\hat{\mu} = X$, NO UMVUE FOR σ^2 EXISTS

SINCE NO UNBIAS ESTIMATE OF σ^2 EXISTS. (HW: SHOW)

EVALUATING ESTIMATORS:

INFORMATION: ONE PARAMETER CASE (593 REVIEW)

REGULAR PARAMETRIC MODEL:

$$\mathcal{F} = \{f_\theta : \theta \in \mathcal{H}\} \quad \mathcal{H} \subset \mathbb{R} \quad X \sim f_\theta$$

UNDER REGULARITY CONDITIONS

$$I(\theta; X) = E_\theta \left[\left\{ \frac{\partial \log f(X; \theta)}{\partial \theta} \right\}^2 \right] \quad \text{MEASURES STABILITY OF } f_\theta \text{ AS } \theta \text{ VARIES}$$

OTHER FORMS: UNDER REGULARITY CONDITIONS

(WHICH ALLOW THE EXCHANGE OF DIFFERENTIATION AND EXPECTATION)

$$\bullet \quad I(\theta; X) = \text{VAR}_\theta \left[\partial \log f(X; \theta) / \partial \theta \right] \quad \text{SINCE } E_\theta[\cdot] = 0.$$

$$\bullet \quad I(\theta; X) = -E_\theta \left[\partial^2 \log f(X; \theta) / \partial \theta^2 \right]$$

ONE-ONE REPARAMETERIZATION: SUPPOSE $\gamma = g(\theta)$ with g being 1-1 and differentiable, then

$$I(\gamma; X) \equiv I(g(\theta); X) = I(\theta; X) / \{g'(\theta)\}^2$$

$$\text{SINCE} \quad \partial \log f(X; \gamma) / \partial \gamma = \frac{\partial \log f(X; g(\theta))}{\partial \theta} \bigg/ \frac{\partial g(\theta)}{\partial \theta},$$

ONE-PARAMETER EXPONENTIAL FAMILY

$$f(x; \eta) = \exp \{ \eta T(x) + d_0(\eta) + S(x) \} I_A(x)$$

$$\Rightarrow \partial \log f / \partial \eta = T(x) + d_0'(\eta) \Rightarrow \partial^2 \log f / \partial \eta^2 = d_0''(\eta)$$

$$\Rightarrow I(\eta; X) = -d_0''(\eta) > 0.$$

FOR AN EXPONENTIAL FAMILY: $\{\eta, T(\underline{x}), d_0(\eta), S(\underline{x})\}$ $I(\eta; \underline{x}) = -d_0''(\eta)$

FOR $\{\theta, T(\underline{x}), d(\theta), S(\underline{x})\}$: $I(\theta; \underline{x}) = I(c(\theta); \underline{x}) \{c'(\theta)\}^2$ $\eta = c(\theta)$.

INFORMATION INEQUALITY: $\text{VAR}_\theta [H(\underline{x})] \geq \{\psi'(\theta)\}^2 / I(\theta; \underline{x})$ $\psi(\theta) = E_\theta [H(\underline{x})]$
 $= 1/I(\psi(\theta); \underline{x})$ IF ψ IS 1-1.

CRAMER-RAO: IF $H(\underline{x})$ IS AN UNBIASED ESTIMATE OF $g(\theta)$, THEN
 $\text{VAR}_\theta (H(\underline{x})) \geq \{g'(\theta)\}^2 / I(\theta; \underline{x})$.

NOTE: ONE-PARAMETER EXPONENTIAL FAMILY. $T(\underline{x})$: NATURAL SUFFICIENT STATISTIC

$$\text{VAR}_\eta (T(\underline{x})) = -d_0''(\eta) \quad \text{AND} \quad E_\eta [T(\underline{x})] = -d_0'(\eta)$$

$$\Rightarrow \text{VAR}_\eta (T(\underline{x})) = \{\psi'(\eta)\}^2 / I(\eta; \underline{x}) \quad \text{I.E. EQUALITY}$$

$$\text{NOTE: } -d_0''(\eta) = d_0''(\eta)^2 / (-d_0''(\eta))$$

THEOREM 5.1, THE INFORMATION INEQUALITY IS AN EQUALITY IFF $H(\underline{x}) = T(\underline{x})$ IS
 the natural sufficient statistic of a $k=1$ parameter EXPONENTIAL FAMILY.

SKETCH OF PROOF. THE KEY STEP IN PROVING THE INFORMATION INEQUALITY IS:

$$\psi'(\theta) = E_\theta \left(H(\underline{x}) \frac{\partial \log f(\underline{x}; \theta)}{\partial \theta} \right) = \text{COV}_\theta \left(H(\underline{x}), \frac{\partial \log f(\underline{x}; \theta)}{\partial \theta} \right)$$

$\Rightarrow$ BY CAUCHY SCHWARTZ

$$\{\psi'(\theta)\}^2 \leq \text{VAR}_\theta (H(\underline{x})) \cdot \text{VAR}_\theta \left(\frac{\partial \log f(\underline{x}; \theta)}{\partial \theta} \right) = \text{VAR}_\theta (H(\underline{x})) I(\theta; \underline{x})$$

So equality holds iff $H(\underline{N})$ and $\frac{\partial \log f(\underline{N}; \theta)}{\partial \theta}$ are linearly dependent

$$\text{i.e. } \frac{\partial \log f(\underline{N}; \theta)}{\partial \theta} = a(\theta) H(\underline{N}) + b(\theta)$$

$$\Rightarrow \log f(\underline{N}; \theta) = A(\theta) H(\underline{N}) + B(\theta) + S(\underline{N}) \quad \square$$

EXAMPLE: $X_1, \dots, X_n$ iid Poisson (λ)

$$f(\underline{N}; \lambda) = \exp \{ (\log \lambda) n \bar{X} - n \lambda + S(\underline{N}) \} \quad A = \{0, 1, \dots, n\}$$

$$\frac{\partial \log f}{\partial \lambda} = \frac{n \bar{X}}{\lambda} \quad n \Rightarrow I(\lambda; \underline{X}) = \text{VAR} \left(\frac{1}{\lambda} \sum X_i \right) = \frac{n}{\lambda^2} \cdot \lambda = \frac{n}{\lambda}$$

NOTE: $E(\bar{X}) = \lambda \Rightarrow \bar{X}$ is MLE of λ (WHY?)

$$\text{ALSO, } \bar{X} \text{ is UMVUE of } \lambda \quad \text{VAR}(\bar{X}) = \frac{1}{I(\lambda; \underline{X})} = \lambda/n$$

$$\text{EXAMPLE: } \gamma = e^\lambda \Rightarrow I(\gamma; \underline{X}) = I(\lambda; \underline{X}) / \gamma'(\lambda)^2 = \frac{n/\lambda}{e^{2\lambda}} = \frac{n}{\gamma^2 \log \gamma}$$

$$\hat{\gamma} = \left(1 + \frac{1}{n}\right)^{n \bar{X}} \text{ is UMVUE for } \gamma \text{ since } E[\hat{\gamma}] = \gamma \quad (\text{HW})$$

$$\text{VAR}(\hat{\gamma}) = e^{2\lambda} (e^{\lambda/n} - 1) = \gamma^2 (\gamma^{1/n} - 1) \quad \text{HW} \quad \text{HENCE } \gamma^2 (\gamma^{1/n} - 1) > \frac{\gamma^2 \log \gamma}{n}$$

$$\text{DIRECT: } \gamma^{1/n} - 1 > \frac{1}{n} \log \gamma \text{ when } \gamma > 1 \quad (\text{HW})$$

$$\text{HOWEVER: } 1 > \frac{\gamma^2 \log \gamma / n}{\gamma^2 (\gamma^{1/n} - 1)} \rightarrow 1 \text{ as } n \rightarrow \infty \quad (\text{HW})$$

DEFINITION 5.2. The FISHER EFFICIENCY of an unbiased estimate $T(\underline{X})$ of $g(\theta)$ is defined to be

$$e(T(\underline{X}); g(\theta)) = \frac{g'(\theta)^2 I(\theta; \underline{X})}{\text{VAR}_\theta(T(\underline{X}))} \leq 1$$

If $X_1, \dots, X_n$ are iid $f(x; \theta)$ and $T_n \equiv T_n(X_1, \dots, X_n)$

IS A CONSISTENT ESTIMATE OF $g(\theta)$, then T_n is said to be

ASYMPTOTICALLY EFFICIENT IF $e(T_n; g(\theta)) \rightarrow 1$ as $n \rightarrow \infty$.

ADDITIVITY OF INFORMATION :

If $X_j \sim f_j(x; \theta)$, $\theta \in (H) \subset \mathbb{R}$, are mutually independent for

$j=1, \dots, k$, then $I(\theta; X_1, \dots, X_k) = \sum_{j=1}^k I(\theta; X_j)$.

PROOF : HW (593 REVIEW)

ASYMPTOTIC PROPERTIES OF MAXIMUM LIKELIHOOD ESTIMATORS

THEOREM 5.3. Suppose $X_1, \dots, X_n$ is a random sample of $X \sim F$ with $F \in \mathcal{F}\{F_\theta; \theta \in (H)\}$, $(H) \subset \mathbb{R}$. Let $\hat{\theta}_n$ represent the MLE of θ , then under certain regularity condition*

$$\sqrt{n} (g(\hat{\theta}_n) - g(\theta)) \rightarrow_d N(0, \sigma_{\theta, g}^2)$$

where $\sigma_{\theta, g}^2 = \{g'(\theta)\}^2 / I(\theta, X)$. For 1-1 g : $\sigma_{\theta, g}^2 = \frac{1}{I(g(\theta); X)}$

* SEE RAO, C.R. (1973). LINEAR STATISTICAL INFERENCE AND ITS APPLICATION, 2ND EDITION. SECTION 5f.2